

2 – משימת הכנה RISC-V האקטון

- Modify the accelerator operation that was presented in the last lecture such that it performs dot product between two 8-element vectors:

- Register Address map:

Address	Register name	Description
0x80001300	Control	Control register
0x80001304	A	Data input vector of 4 elements
0x80001308	B	Data input vector of 4 elements
0x8000130C	C	Data input vector of 4 elements
0x80001310	D	Data input vector of 4 elements
0x80001314	Result	Result

- The Control register has:
 - A "Go" bit on its least significant bit (bit 0). Writing 1 to the Go bit will activate the accelerators.
 - A "done" bit on its most significant bit (1 indicate that results is ready).
- The accelerator will calculate the dot-product as follow:
 - $\text{Result} = A[3]*B[3]+A[2]*B[2]+A[1]*B[1]+A[0]*B[0]+C[3]*D[3]+C[2]*D[2]+C[1]*D[1]+C[0]*D[0]$
- Write a C program that demonstrates the accelerator operation by performing matrix multiplication on the following 8x8 matrices:

[[53, 24, 92, 10, 49, 99, 15, 72], [30, 69, 88, 77, 45, 65, 16, 81], [24, 6, 55, 26, 31, 48, 37, 12], [4, 71, 98, 82, 59, 39, 79, 97], [96, 40, 14, 95, 50, 29, 66, 17], [2, 94, 33, 61, 91, 88, 78, 54], [60, 35, 73, 62, 57, 18, 47, 85], [20, 11, 21, 83, 38, 44, 64, 91]]	[[30, 66, 84, 88, 13, 49, 19, 93], [28, 31, 67, 14, 79, 17, 27, 45], [50, 56, 47, 24, 34, 57, 70, 82], [8, 60, 54, 76, 80, 33, 40, 95], [4, 75, 22, 29, 59, 41, 64, 29], [95, 12, 71, 53, 14, 38, 22, 16], [7, 37, 68, 41, 61, 12, 58, 55], [99, 65, 25, 20, 46, 91, 32, 68]]
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The C program should print the result matrix

- Note the result matrix should be:
[[15291, 20427, 20203, 18284, 16865, 18476, 19621, 24439],
[18453, 21030, 24333, 20801, 21935, 20557, 20711, 25766],
[9616, 11402, 12381, 10922, 10970, 10189, 11207, 13295],
[18682, 22467, 26138, 22023, 24368, 21863, 25551, 29722],
[15583, 18683, 19210, 18690, 19493, 17219, 17811, 21456],
[21211, 23045, 27089, 24009, 24890, 22234, 24290, 28472],
[15599, 18586, 20257, 18570, 19346, 17418, 18588, 22271],
[17217, 19985, 21918, 19380, 19975, 18323, 18826, 23037]]

- Submission:
 - The new modified accelerator system_verilog files:
 - accelerator_top.sv
 - accelerator_wb.sv
 - accelerator_regs.sv
 - accelerator.sv
 - The new modified C program that demonstrates the operation of the accelerator.
 - A waveform snapshot from Vivado simulation demonstrating the operation of the accelerator.
 - A waveform snapshot from FPGA run using iLA demonstrating the operation of the accelerator.
 - A screen snapshot of the output result